Chapter 1

INTRODUCTION AND BACKGROUND

Modern manufacturing relies heavily on producing precise, intricate metal components. This is often achieved through subtractive manufacturing techniques using equipment such as milling machines and lathes. These Machines have been in use for approximately two hundred years[Groover and Moriwaki], and during this time have been continually improved. Manual control of these Machines has incrementally replaced by increasingly capable computerization [Reintjes, Groover]. Modern milling Machines and lathes often use Computer Numerical Control (CNC), which removes the need for a human machinist to control each detail of the machining process through manual inputs. Due to this continued computerization, these machines have grown in complexity, enabling more advanced machining operations, including advanced work with ductile materials, brittle materials, and increasing complex part geometry. [Civiello, Groover] Continued development has also introduced new systems to these Machines, including systems that allow the Machine to autonomously detect in real time a broken tool and replace the tool, detect abnormally high stress loads, or monitor motor loads across the machine. These innovations are significant because they enable the Machines to autonomously catch in real time errors that previously would have been caught by a human, and with CNC Machines could have propagated into damaged parts or damaged Machines. These innovations work to reduce scrap part rates, damaged tooling, and critically machine downtime.

Lathe

A lathe performs subtractive manufacturing by rotating the workpiece around the spindle axis, designated as the Z axis, while translating the cutting tool in the X and Z axes. The X axis is commonly used to move the tool radially and controls the finished diameter of the workpiece. The Z axis moves along the length of the workpiece and is used to control the axial extent of the cut. Given the workpiece rotates while the tool traverses a plane containing the axis of rotation, all features are produced is a surface revolution and axially symmetrical about the Z axis. [Citation needed for the standard lathe coordinate system and machining structure] Lathes are commonly used to achieve part geometries that require turning outer diameters, boring internal diameters, facing, and drilling and tapping holes along the Z axis. Contoured profiles, including hemispherical and other curved forms are produced by coordinating the X and Z motions simultaneously. A conventional lathe has no Y axis, the Y axis is traditionally used only on mill-turn machines and multi-axis turning centers. The Y axis enables the off axis of rotation features which would break axial symmetry. [Citation needed for the Y axis in a lathe].

During turning passes, the tool remains engaged with the workpiece continuously, therefore unless in absence of material variation or progressive tool wear, the cutting force is approximately consistent. This force is conventionally resolved into three orthogonal components, tangential, radial, and feed. [citation needed for force directions] The magnitude of each of these forces is determined largely by the tool geometry, machine geometry, and the part characteristics. Large contributions to this are determined by the tool geometry and the lead and rake angles. [Citation needed for the load vectors and definitions]

Mill

A milling machine performs subtractive manufacturing by rotating the cutting tool about the spindle axis, designated the Z axis in a milling machine. Part geometry is achieved by moving the tool and workpiece relative to each other along three orthogonal directions. On a standard vertical machining center, the table carries the workpiece in the X and Y axes while the spindle head provides motion in the Z axis setting the depth of cut. Through these three degrees of freedom, material can be removed as arbitrary locations on the workpiece, so a far wider range of geometry can be produced compared to a lathe. Additional rotary axes extend this part geometry further. Four-axis machines add a single rotary axis about the X or Y axis and five-axis machines add a second rotary axis allowing the tool to approach the workpiece at a continuously variable orientation and make undercut and complex contoured surfaces without rotational symmetry possible. [Axis definitions and part geometry citation needed]

The cutting force in a milling operation is applied to the workpiece through the flutes of the rotating tool, and each flute enters and leaves the cut once per spindle rotation. Engagement therefore is intermittent rather than continuous, and the load present on the machine is periodic at the tooth-passing frequency. The tooth-passing frequency is equal to the spindle rotational speed multiplied by the number of flutes on the cutting tool. [Citation needed for the rotational properties, endmill definitions, and intermittent load characteristics]

CNC Based Manufacturing

Both lathes and mills can be improved by the addition of computer based control, and machines of this type are known as Computer Numerical Control (CNC) lathes and mills. Through CNC, tool paths are specified as coordinated motion commands rather than performed by an operator adjusting hand-wheels and cranks, CNC toolpaths allow for simultaneous

coordinated multi-axis motions that are required for complex contoured geometry that would not be practical when performed by a human operator using hand-wheels and cranks to control the machine. [Citation needed regarding the additional complex geometries enabled by CNC] On servo based machines, motion is regulated by nested closed-loop position and velocity controllers acting in each axis [Citation needed regarding modern control mechanisms for servo based machines] Because the controller compares the commanded position against measured position at each update of the servo loop, often occurring faster than the machine could reach the desired location, position error is corrected for continuously and is therefore corrected. This control system allows CNC machines to hold a high level of precision and to reproduce the same result across many parts. Machines driven by stepper motors are commonly used and typically operated in an open-loop configuration. This principle takes advantage of the internal stepping and gearing motions a stepper motor provides to achieve reproducible and precise control at a lower level than servo based machines, but at a lower cost and reduced complexity. [Citation needed about different drive types, especially the difference in performance of stepper based control.]

Adding sensing beyond the position control loop permits tighter control and the recording of process data during machining. This data can be retained on a per-part and per-machine basis and support process improvements across a production run in a manner that manual machining did not allow for. [Citation on data collection and process manufacturing improvements made to machines]

High-performance CNC machines are typically expensive and occupy large floor footprints. These large often cast based footprints are used to raise the mass and structural dampening of the machine, attenuating vibration and disturbance forces before tool disturbance can occur. This contributes substantially to the precision and repeatability these machines can achieve [Citation regarding machine vibrations and precision needed] If these disturbances can

instead by characterized and rejected within the control system, a portion of the precision obtained by this structural mass could in principle be obtained form control, allowing smaller and lighter machines to approach the performance of larger cast based machines. Machines of this type would be usable in applications where the machine must be enclosed and where cost or space required for this enclosure currently constrains what can be produced.

Observability of Machining Loads

Some modern CNC machines are capable of recording substantial data during operation, and that data is presently used in three primary ways: to improve part quality across a production run, to reduce machine cycle time, and to indicate when maintenance should be performed. Each of these acts after the fact, as the part being cut when the data was collected is already complete. With additional sensing and characterization of the machine, the same information could support correction during the current cut, acting on the workpiece currently being machining rather than the next one. Corrections of this kind would also accumulate, producing a record of which in-process adjustments succeeded which could be applied to a new part geometry without repeating the same trial cuts.

Systems performing monitoring of this kind are already commercially available. The Tool Monitoring Adaptive Control (TMAC), developed by Caron Engineering interfaces directly with a range of CNC controls, including Okuma platforms. The system monitors spindle motor power during the cut and operates on the principle that the power required to cut increases as the cutting edge wears. It applies this logic in two ways, it retires a worn tool or halts the machine on breakage, and in adaptive control mode, it overrides the programmed feed rate to hold spindle motor power near a set limit. [Cite Sam’s work on TMAC] Data from these cuts can be retained for later analysis and process improvements. [Cite Sam’s work again]

The adaptive capability therefore exists, but it acts on an aggregate quantity. Spindle motor power represents the total power drawn by the cut. It does not resolve the force at the cutting edge into its principle components, and holding that power near a limit regulates the process only in the coarsest sense, without reference to the cutting conditions that have produced these changes. [Cite Sam's thesis] Through the work completed in, [Sam's thesis source number] found that cutting force at the cutting edge could not be resolved from the data accurately enough to infer the resulting change in part geometry or surface finish. This limitation was attributed to the sample rate of the TMAC system, where the noise was no longer distinguishable from the signal. The signal was sampled at a rate of 200Hz which was well below the floor of the noise. This resulted in the data that showed the process was feasible, however the electrical sampling system would need significant changes to be practical. A system with direct access to the motor's electrical quantities and closer integration with the motor may resolve the signal to noise ration problem and provide more insight than the TMAC system could.

With advances in modern CNC manufacturing equipment, new data can be collected. Currently this data can be used for process improvements to improve part quality over the life cycle of a part, decrease machine run time for part completion, or to infer when maintenance should be completed. With additional sensing and machine characterization, in situ changes could be made to prevent the current workpiece from failure rather than waiting for part completion to make process improvements on the large scale. This focus on improving the current part can be used to build up a smaller scale of improvements that happen live eventually building to a repertoire of changes that can be referenced on new part geometry.

Machining Load Transmission

When an axis is driven by a ballscrew, the cutting force acting on the carriage produces an axial load on the ballscrew. Given that a ballscrew is backdrivable the axial load appears at

the motor as a rotational torque. The magnitude of this torque is set by the screw lead and the backdriving efficiency of the drive. [Machine design/Ballscrew dynamics citation needed] The cutting load therefore reaches the motor through the existing drive system, without additional instrumentation added to the machine. Many CNC machines come standard with ballscrews, however those that use alternative drive types such as lead screws can be retrofitted with ballscrews to achieve this behavior and enable this torque transmission.

Electric motors produce torque as a function of supplied current, the form of this current torque relationship varies on motor design. However, all electric motors fundamentally convert electrical current into a magnetic field that is used to produce torque. Some motor types produce torque at a fixed linear relationship to current, while other are inferred through a non-linear relationship dependent on rotor angle. Determining the load from the current therefore requires a priori knowledge on the electrical characteristics of the motor. This torque to current relationship motivates the review of motor types in the following section.

Where the relationship is known, the cutting load appears at the axis drive as a disturbance torque, and the controller responds by changing the command current in order to preserve the axis trajectory. The cutting load is consequently present in the current signal and can be recovered from this disturbance. Recovery of this cutting load requires separating this load from the other torques the motor supplies during the same motion. These additional disturbance torques include friction in the system, inertial torque required to accelerate and decelerate the axis, and on a vertical axis the effects of gravity on the load of the carriage. Inertial and gravitational contributions follow from the axis mass and command trajectory can be predicted directly and therefore identified as a known disturbance. Friction is less traceable, comprising Coulomb, viscous, and velocity dependent component that vary with temperature and exhibit a hysteresis at velocity change. [Citation needed for friction characteristics and ballscrew motor dynamics] These disturbance therefore instead of being predicted are instead estimated and

corrected for in situ through observer-based and signal processing techniques present in the proposed control system. [Citation needed for friction estimation with observers and the ADRC]

The underlying principle for torque based disturbance characterization is already applied in a limited form on modern CNC machines. This is used where drive current detects events without any added sensor. These methods are commonly described as “sensorless”. The term originates in motor control, where it denotes the estimation of rotor position without a position sensor. In this context is is used in the broader sense adopted in machine tool practice, meaning that the quantity of interest is inferred from signals already present in the drive rather than measured by a sensor added for that purpose. Homing against a hard stop is one example. When the axis reaches the stop, the motion ceases while the controller continues to the command position. This position error accumulates and the commanded current rises toward the motor limit. This sustained rise identifies a hard stop and can be used to establish a machine reference point without a limit switch. [Citation needed for this technology. Used in 3D printer and specifically BambuLabs and Voron printers] This hard stop referencing method is only practical due to the relatively high speeds at which the controller is able to sample error and detect a disturbance so no damage to the machine occurs during this brief period of time when the hard stop is in contact with the carriage on that axis.

This principle can be expanded to detect other machine changes, tool breakage is another common use case for detecting sudden current load changes. This is normally performed by monitoring the spindle drive rather than an axis drive. The sudden loss of a cutting edge produces an abrupt decrease in spindle load, most commonly a decrease in power as the tool is no longer in contact with the workpiece. [Citation for tool breakage detection needed] This signal is drawn from a different motor, however the principle is the same in both use cases. Both the axial hard stop and the tool breakage are determined by sudden change in current draw in the

motor controlling that motion. The present work extends this same principle from an event detection approach to a continuous estimation of the load magnitude.

Characterizing Load at the Cutting Edge

With a motor and control system of sufficient electrical and control bandwidth, current can be sampled rapidly enough to follow variation in the cutting load rather than only tracking an average value. Turning is the appropriate operation in which to establish this relationship first. The tool remains engaged continuously through a turning pass, and during a hemispherical operation translates in both the X and Z axes causing a change in the force vector direction. With these passes, in the absence of material variation or progressive tool wear, the cutting force is approximately steady and the motor torque is likewise steady. Milling by contrast, presents an intermittent load at the tooth-passing frequency, superimposing a large period component on the signal of interest. Beginning with the turning isolates the current-to-force relationship from the periodic relationship which allows it to be established under the simplest available loading condition.

Direct instrumentation of the cutting zone is not viable in the target applications, where the machining environment is enclosed and access is limited. Introducing sensors into this environment introduces a non-inconsequential cost aspect and would expose sensors to potential damage due to conditions internal to the machine during operation. Using already existing mechanics to inter the load from the motor current instead, locates the measurement at the drive, where the necessary electrical quantities are already present in the controller, and requires no instrumentation in the cutting zone at all.

Establishing this relationship also opens a path to quantity that is not otherwise observable during machining. Force at the cutting edge is distributed over the contact between the tool and the workpiece, and the pressure in that contact, rather than the total force, is what

governs the material's response to the cut. Pressure follows from force and the area over which it acts, and that area is determined by the chip cross-section and geometry of the cutting tool. [Citation for chip formation, and Sam's work on pressure to cutting force] Specific cutting energy provides the standard basis for relating the measured force to the condition in the cutting zone. [Cite Sam's thesis again] Where force can be recovered from drive current, tool geometry, and chip cross-section from machining parameters, an estimate of pressure at the cutting edge can be determined without additional measurement.

A pressure estimate obtained through this could be of use where a material's machining envelope depends on the loading conditions it experiences rather than on nominal properties alone. Certain material classes undergo solid-state phase transforms within the range of stress and temperature encountered at the cutting edge, and for these deformations the behavior is dependent on the pressure and temperature history the material has experienced. [Citation needed for manufacturing HE] Machining such materials predictably required a priori knowledge of the load actually being applied at the cutting edge, which no commercial machine presently provides.

Recovering pressure in this manner is beyond the scope of the present work, which establishes a relationship between motor current and axis load that any such estimate would depend upon. This approach is identified here because it defines the direction in which this capability could extend, and because the requirements this imposes depend on the method of data collection in the present work. This includes, bandwidth sufficient to follow the cut, a torque-to-current relationship, and a control system capable of accepting current feedback. Meeting these requirements on a stepper-driven axis requires a motor model, a control architecture capable of maintaining the torque-to-current relationship under load, and a processing platform capable of operating at the required rate. Each of which is developed in the present work.